1 **(a)** $\frac{24}{60} \times 100$ **[M1]**

40 **[A1]**

(b) $\frac{15}{100} \times 80 = 12$ OR 80×0.85 **[M1]**

68 **[A1]**

[Total Marks: 4]

2 **(a)** Discount $= 0.15 \times 120 = 18$ OR 0.85×120 **[M1]**

102 **[A1]**

(b) Let cost price be C. $1.2C = 120$ **[M1]**

$C = \frac{120}{1.2}$ **[M1]**

100 **[A1]**

[Total Marks: 5]

3 **(a)** $45\,000 \times 1.042$ OR $45\,000 + 45\,000 \times 0.042$ **[M1]**

46 890 **[A1]**

(b) $100\% - 24\% = 76\%$ **[B1]**

$\frac{342}{0.76}$ OR $\frac{342}{76} \times 100$ **[M1]**

450 **[A1]**

[Total Marks: 5]

4 **(a)** $5000 \times \left(1 + \frac{2.5}{100}\right)^3$ **[M1]**

5384.45 **[A1]**

(b) $\left(1 + \frac{r}{100}\right)^5 = 1.30$ **[M1]**

$1 + \frac{r}{100} = \sqrt[5]{1.30}$ **[M1]**

5.39 (or $5.387\ldots$) **[A1]**

(c) $V \times (1 - 0.12)^3 = 17\,036.80$ **[M1]**

$V = \frac{17\,036.80}{0.88^3}$ **[M1]**

25 000 **[A1]**

[Total Marks: 8]

5 **(a)** $4500 \times (1 + \frac{2.4}{100})^5$ or 4500×1.024^5 **[M2]**

If M0, M1 for 4500×1.024 or 4608 seen.

5066.57 or $5066.56\ldots$ **[A1]**

(b) Interest earned $= 5066.57 - 4500 = 566.57$ (accept unrounded) **[M1]**

$\dfrac{4500 \times r \times 5}{100} = 566.57$ OR $225r = 566.57$ **[M1]**

2.52 or $2.518\ldots$ **[A1]**

[Total Marks: 6]

6 (a) $1\,250\,000 \times (1 + \dfrac{x}{100})^3 = 1\,367\,631$ **[M1]**

$(1 + \dfrac{x}{100})^3 = 1.0941048$ **[M1]**

$1 + \dfrac{x}{100} = \sqrt[3]{1.0941048} = 1.0304\cdots \implies x = 3.04\cdots \approx 3$ **[A1]**

(b) Target population $= 2 \times 1\,250\,000 = 2\,500\,000$ **[B1]**

$1\,250\,000 \times (1.03)^n > 2\,500\,000 \implies 1.03^n > 2$ **[M1]**

Using logs: $n\log(1.03) > \log(2)$ OR trial and improvement yielding n between 23 and 24 **[M1]**

$n > 23.449\ldots$, so it takes 24 years. Year $= 2015 + 24 = 2039$ **[A1]**

[Total Marks: 7]